

Table 3-7 defines the full set of possible wires needed to produce USB Type-C to legacy cable assemblies. For some cable assemblies, not all of these wires are needed. For example, a USB Type-C to [USB 2.0](#) Standard-B cable will not include wires 5–10.

Table 3-7 USB Type-C Cable Wire Assignments for Legacy Cables/Adapters

Wire Number	Signal Name	Description
1	GND_PWRrt1	Ground for power return
2	PWR_VBUS1	VBUS power
3	UTP_Dp	Unshielded twist pair, positive
4	UTP_Dn	Unshielded twist pair, negative
5	SDPp1	Shielded differential pair #1, positive
6	SDPn1	Shielded differential pair #1, negative
7	SDP1_Drain	Drain wire for SDPp1 and SDPn1
8	SDPp2	Shielded differential pair #2, positive
9	SDPn2	Shielded differential pair #2, negative
10	SDP2_Drain	Drain wire for SDPp2 and SDPn2
Braid	Shield	Cable external braid

Note:

- a. This table is based on the assumption that shielded twisted pair is used for all SDP's and there are drain wires. If coaxial wire construction is used, then no drain wires are needed and the signal ground return is through the shields of the coaxial wires.

3.3.3 Wire Gauges and Cable Diameters (Informative)

This specification does not specify wire gauge. Table 3-8 and Table 3-9 list typical wire gauges for reference purposes only. A large gauge wire incurs less loss, but at the cost of cable diameter and flexibility. Multiple wires may be used for a single wire such as for VBUS or Ground. It is recommended to use the smallest possible wire gauges that meet the cable assembly electrical and mechanical requirements.

To maximize cable flexibility, all wires should be stranded and the cable outer diameter should be minimized as much as possible. A typical USB Full-Featured Type-C cable outer diameter may range from 4 mm to 6 mm while a typical [USB 2.0](#) Type-C cable outer diameter may range from 2 mm to 4 mm. A typical USB Type-C to [USB 3.1](#) legacy cable outer diameter may range from 3 mm to 5 mm.